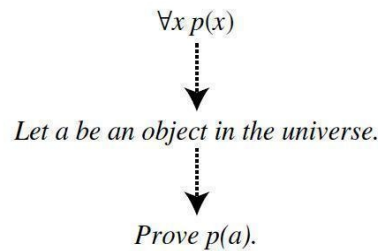


Universal Proofs

To prove $\forall x p(x)$ from a given set of premises, we show that every object satisfies $p(x)$ assuming those premises. Since the proofs are mathematical, we can restrict the objects to a particular **universe**. A proposition of the form $\forall x p(x)$ is then interpreted to mean that $p(x)$ holds *for all x from a given universe*.



These types of proofs are called **universal proofs**.

Note:

Definition:

For all integers a and b , a **divides** b (written as $a \mid b$) if $a \neq 0$ and there exists an integer k such that $b = ak$.

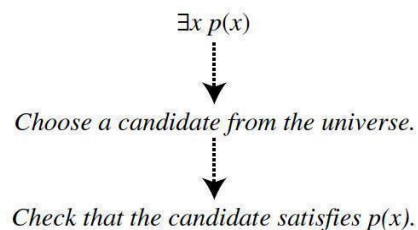
Definition:

An integer n is **even** if $2 \mid n$, and n is **odd** if $2 \mid (n - 1)$.

Existential Proofs

Suppose that we want to write a proof for $\exists x p(x)$. This means that we must show that there exists at least one object of the universe that satisfies the formula $p(x)$.

It will be our job to find that object. To do this directly, we pick an object that we think will satisfy $p(x)$. This object is called a **candidate**. We then check that it does satisfy $p(x)$. This type of a proof is called a **direct existential proof**, and its structure is illustrated as follows:



Existence and Uniqueness

There will be times when we want to show that there exists exactly one object that satisfies a given predicate $p(x)$. In other words, there exists a **unique** object that satisfies $p(x)$. This is a two-step process.

- **Existence:** Show that there is at least one object that satisfies $p(x)$.
- **Uniqueness:** Show that there is at most one object that satisfies $p(x)$. This is usually done by assuming both a and b satisfy $p(x)$ and then proving $a = b$.

Note:

The phrase *exactly one* occurs so often in mathematics that there is a special notation for it. We will write $\exists! x P(x)$ to represent the statement

“There is exactly one value of x such that $P(x)$ is true.”

It is sometimes also read “There is a unique x such that $P(x)$.”

For example, the $2x + 1 = 5$ has a unique real solution.

i.e. $\exists! x \in \mathbb{R}, 2x + 1 = 5$.

Direct proofs

This gives a fundamental outline for proving statements of the form $P \Rightarrow Q$. Begin by assuming that P is true (remember, we don't need to worry about P being false) and show this forces Q to be true.

Outline for Direct Proof

Proposition If P , then Q .

Proof. Suppose P .

$\vdots$

Therefore Q . ■

Direct proof of " $\forall x \in D, P(x) \Rightarrow Q(x)$ ":

Let $x \in D$ -----(i)

Suppose that $P(x)$. -----(ii)

$\left. \begin{array}{l} - \\ - \\ - \\ - \end{array} \right\} \begin{array}{l} \text{Definitions,} \\ \text{Axioms,} \\ \text{Logical equivalences,} \\ \text{Rules of inference} \end{array}$

Therefore, $Q(x)$. -----(iii)

By (i), (ii), (iii) we get $\forall x \in D, P(x) \Rightarrow Q(x)$. ■

Important Connection

The biconditional itself expresses logical equivalence:

$P \equiv Q$ means $P \Leftrightarrow Q$ is always true

So:

- " P and Q are logically equivalent"
means
- " $P \Leftrightarrow Q$ " is a tautology.

Example:

$$P \Rightarrow Q \equiv \neg P \vee Q$$

because

$$(P \Rightarrow Q) \Leftrightarrow (\neg P \vee Q)$$

is always true.

Proof by Cases

The last type of proof that relies on direct proof is proof by cases. Suppose that we want to prove $p \Rightarrow q$ and this is difficult for some reason. We notice, however, that p can be broken into cases. Namely, there exist $p_0, p_1, \dots, p_{n-1}$ such that

$$p \Leftrightarrow p_0 \vee p_1 \vee \dots \vee p_{n-1}. \quad p \equiv (p_0 \vee p_1 \vee \dots \vee p_{n-1})$$

If we can prove $p_i \Rightarrow q$ for each i , we have proved $p \Rightarrow q$. If $n = 2$, then $p \Leftrightarrow p_0 \vee p_1$, and the justification of this is as follows:

$$\begin{aligned} (p_0 \Rightarrow q) \wedge (p_1 \Rightarrow q) &\Leftrightarrow (\neg p_0 \vee q) \wedge (\neg p_1 \vee q) \\ &\Leftrightarrow (q \vee \neg p_0) \wedge (q \vee \neg p_1) \\ &\Leftrightarrow q \vee \neg p_0 \wedge \neg p_1 \\ &\Leftrightarrow \neg p_0 \wedge \neg p_1 \vee q \\ &\Leftrightarrow \neg(p_0 \vee p_1) \vee q \\ &\Leftrightarrow (p_0 \vee p_1) \Rightarrow q \\ &\Leftrightarrow p \Rightarrow q. \end{aligned}$$

$$p \equiv (p_0 \vee p_1)$$

Inference Rule (Proof by Cases)

For every positive integer n , if $p \iff p_0 \vee p_1 \vee \cdots \vee p_{n-1}$, then

$$p_0 \implies q, p_1 \implies q, \dots, p_{n-1} \implies q \iff p \implies q.$$

For example, since a is a real number if and only if $a > 0$, $a = 0$, or $a < 0$, if we needed to prove a proposition about an arbitrary real number, it would suffice to prove the result individually for $a > 0$, $a = 0$, and $a < 0$.

For example, n is an integer **if and only if** n is an even integer **or** n is an odd integer

Example 1: If $n \in \mathbb{N}$, then $1 + (-1)^n(2n - 1)$ is a multiple of 4.

Proof. Suppose $n \in \mathbb{N}$.

Then n is either even or odd. Let's consider these two cases separately.

Case 1. Suppose n is even. Then $n = 2k$ for some $k \in \mathbb{Z}$, and $(-1)^n = 1$.

Thus $1 + (-1)^n(2n - 1) = 1 + (1)(2 \cdot 2k - 1) = 4k$, which is a multiple of 4.

Case 2. Suppose n is odd. Then $n = 2k + 1$ for some $k \in \mathbb{Z}$, and $(-1)^n = -1$.

Thus $1 + (-1)^n(2n - 1) = 1 - (2(2k + 1) - 1) = -4k$, which is a multiple of 4.

These cases show that $1 + (-1)^n(2n - 1)$ is always a multiple of 4.

Therefore, if $n \in \mathbb{N}$, then $1 + (-1)^n(2n - 1)$ is a multiple of 4. ■

Example 2: If two integers have opposite parity, then their sum is odd.

Proof. Suppose m and n are two integers with opposite parity.

We need to show that $m + n$ is odd. This is done in two cases, as follows.

Case 1. Suppose m is even and n is odd.

Case 2. Suppose m is odd and n is even.

Note:

The two cases in this proof are entirely alike except for the order in which the even and odd terms occur. It is entirely appropriate to just do one case and indicate that the other case is nearly identical. The phrase “*Without loss of generality...*” is a common way of signaling that the proof is treating just one of several nearly identical cases. Here is a second version of the above example.

Example 2: If two integers have opposite parity, then their sum is odd.

Proof. Suppose m and n are two integers with opposite parity.

We need to show that $m + n$ is odd.

Without loss of generality, suppose m is even and n is odd.

Thus $m = 2a$ and $n = 2b + 1$ for some integers a and b .

Therefore $m + n = 2a + 2b + 1 = 2(a + b) + 1$, which is odd ■

Note:

In reading proofs in other texts, you may sometimes see the phrase “Without loss of generality” abbreviated as “WLOG.” However, in the interest of transparency we will avoid writing it this way. In a similar spirit, it is advisable—at least until you become more experienced in proof writing—that you write out all cases, no matter how similar they appear to be.

Ex1: If two integers have opposite parity, then their product is even.

Ex2. if n is a natural number, then $n^2 + n + 3$ is odd.

Ex3. For each integer n , $n^2 - 5n + 7$ is an odd integer.

Ans EX2. Suppose n is a natural number — (i)
case (i) n is even:

A proof by contraposition or contrapositive proof

Since $P \Rightarrow Q$ is logically equivalent to $\sim Q \Rightarrow \sim P$, it follows that to prove $P \Rightarrow Q$ is true, it suffices to instead prove that $\sim Q \Rightarrow \sim P$ is true. If we were to use direct proof to show $\sim Q \Rightarrow \sim P$ is true, we would assume $\sim Q$ is true and use this to deduce that $\sim P$ is true. This in fact is the basic approach of contrapositive proof, summarized as follows.

Outline for Contrapositive Proof

Proposition If P , then Q . $\equiv$	$\sim Q \Rightarrow \sim P$
<i>Proof.</i> <u>Suppose $\sim Q$.</u>	
$\vdots$	
Therefore $\sim P$. ■	

So the setup for contrapositive proof is very simple. The first line of the proof is the sentence “*Suppose Q is not true.*” (Or something to that effect.) The last line is the sentence “*Therefore P is not true.*” Between the first and last line we use logic and definitions to transform the statement $\sim Q$ to the statement $\sim P$.

Examples:

- (i) Let m be an integer. Prove that if m^2 is even, then m is even.
- (ii) For each integer n , if n is an odd integer, then n^2 is an odd integer.

Ex 1: Suppose $x \in \mathbb{Z}$. If $7x + 9$ is even, then x is odd.

Ex 2: Suppose $x \in \mathbb{Z}$. If $x^2 - 6x + 5$ is even, then x is odd.

Ex 3: Suppose $x, y \in \mathbb{R}$. If $y^3 + yx^2 \leq x^3 + xy^2$, then $y \leq x$.

Ex 4: Suppose $x, y \in \mathbb{Z}$. If $5 \nmid xy$, then $5 \nmid x$ and $5 \nmid y$.

} Ex

Note:

Indirect proofs

- Consider an implication: $p \Rightarrow q$
- It's contrapositive is $\sim q \Rightarrow \sim p$ is logically equivalent to the original implication.
- Thus, show that if $\sim q$ is true, then $\sim p$ is true.
- To perform an indirect proof, do a direct proof on the contrapositive.

Proof.

Assume $\sim Q$.

$\vdots$

Therefore, $\sim P$.

Thus, $\sim Q \Rightarrow \sim P$

Therefore, $P \Rightarrow Q$. ■

End of Pg